

Assignment-1.5

“PC Practicals”

TASK-1

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

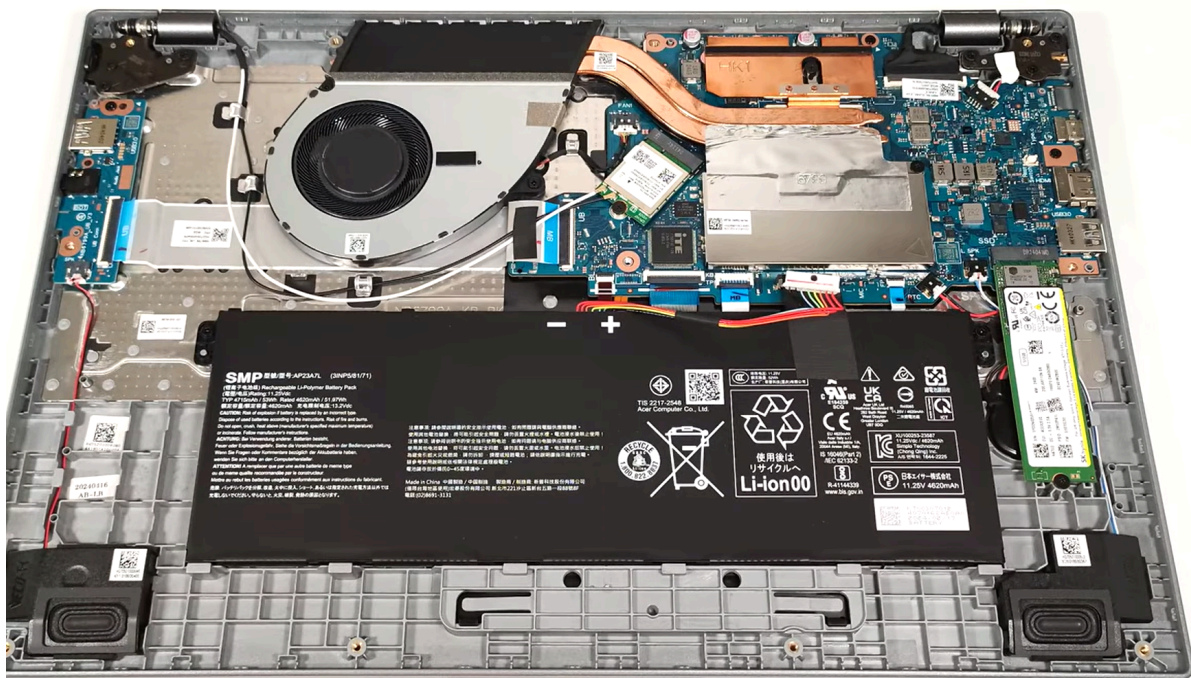
- **Motherboard-** The primary circuit board that connects all computer components (CPU, RAM, storage, expansion cards) together and allows them to communicate. It routes power from the PSU to various parts and acts as the central backbone for system data flow.
- **Central Processing Unit (CPU)-** Known as the brain of the computer, it executes instructions, handles logic, and performs calculations for the system. It processes all tasks given by the operating system and applications to ensure smooth operation.
- **CPU Cooler / Heat Sink & Fan-** A thermal fan placed directly over the CPU to prevent overheating during operation. It draws intense heat away from the processor chip using thermal paste, metal fins, and an active fan.
- **Random Access Memory (RAM)-** High-speed, volatile short-term memory that holds active data and running applications for quick processor access. It temporarily stores files your computer needs right now, but clears entirely whenever the system is powered off.
- **Power Supply Unit (PSU)/SMPS-** Converts high-voltage AC electrical power from the wall outlet into regulated low-voltage DC power for internal components. It distributes specific voltage cables directly to the motherboard, drives, and dedicated graphics card.
- **Storage Drives (HDD / SSD / NVMe)-** Non-volatile storage devices responsible for permanently holding the operating system, installed applications, and user files. Unlike RAM, these drives safely retain all your stored data even after the computer is turned completely off.
- **Graphics Processing Unit (GPU / Expansion Card)-:** A dedicated expansion card fitted into a PCIe slot designed to process graphical data and render output to displays. It offloads visual processing tasks from the main CPU to handle heavy 3D gaming, video editing, and rendering.

- **Cabinet / Case Fans:** Internal fans mounted on the cabinet walls to establish proper directional airflow and exhaust heat out of the case. They draw cool air inside from the front/bottom and push warm ambient air out through the rear/top vents.
- **SATA Cables-** It carries data between data storage devices like SSDs /HDDs and also motherboards.
- **Expansion card-** we can install additional optional into the motherboard slot to add features like wifi or sound.

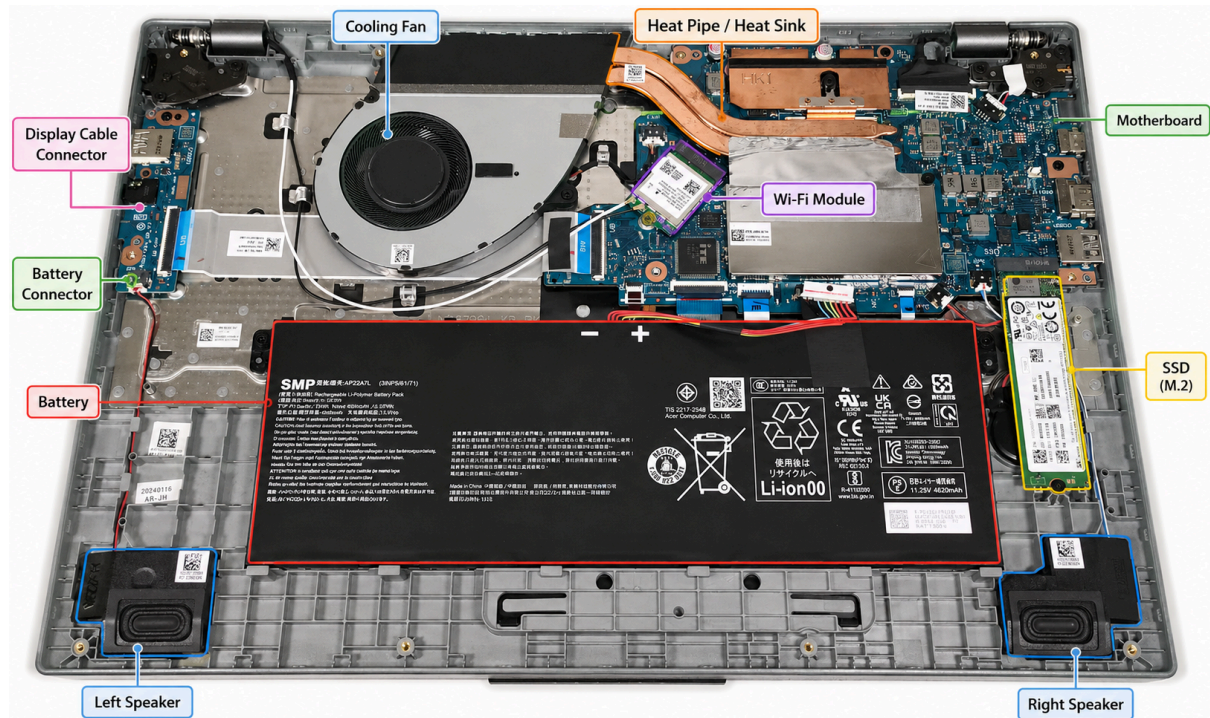
TASK-2

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

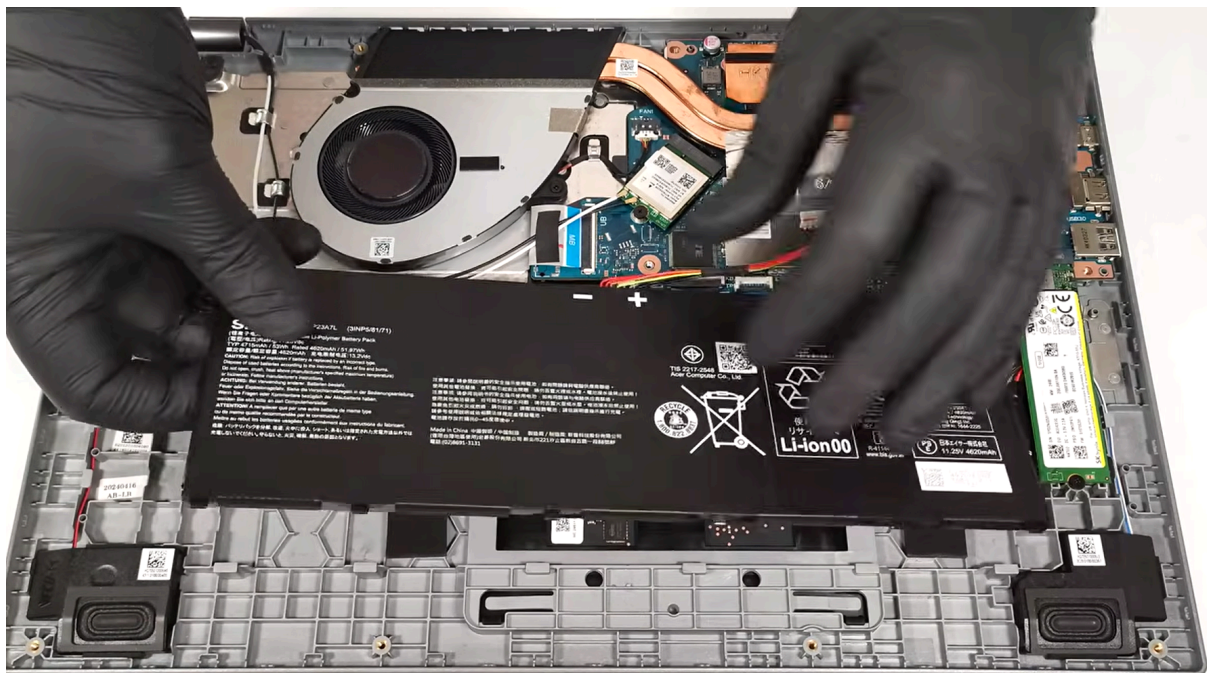
STEP 1: first i shutdown the Laptop and opened bottom cover with screwdriver.



STEP 2: labelling all components.



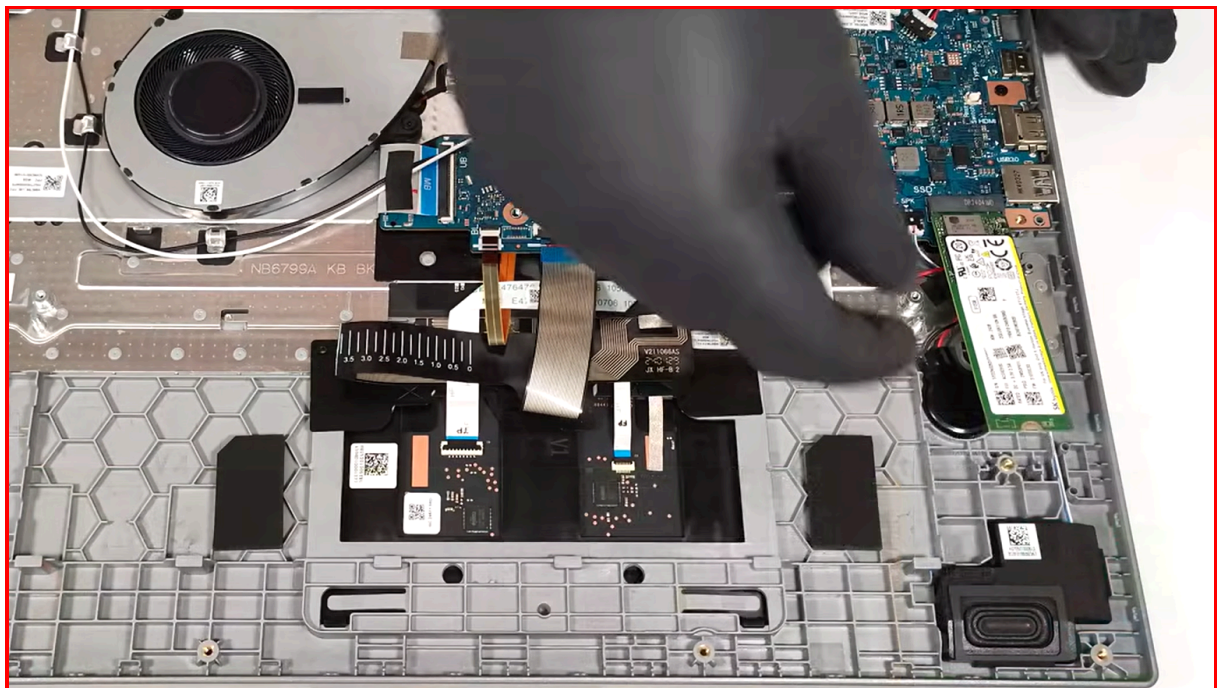
STEP 3: removing battery



STEP 4: released the clips and removed the RAM module.



STEP 5: removing SSD.



STEP 6: laptop is now disassembled and all main components are visible and well organised for reassemble.

TASK-3

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step 1 — Reinstall the internal components

Placing the internal components back into their original positions

Why:

Correct positioning prevents components from being damaged and ensures that the laptop can be closed properly.



Step 2 — Install the M.2 SSD

Installing and securing the M.2 SSD

Why:

The SSD provides permanent storage for the operating system, applications and personal files.

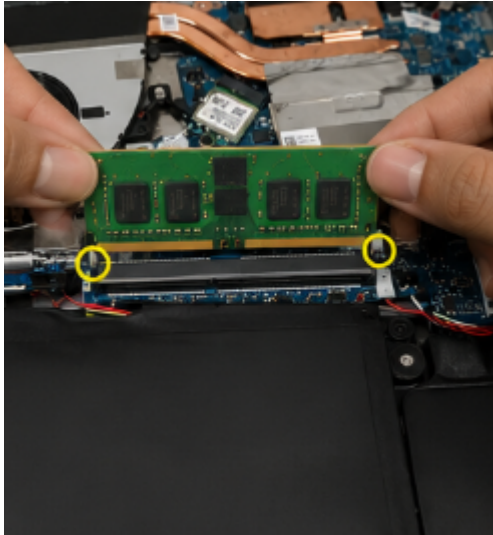


Step 3 — Reinstall the RAM

Reinstalling the RAM module

Why:

RAM must be properly seated so the laptop can access its temporary working memory.



Step 4 — Reconnect the battery

Reconnecting the laptop battery to the motherboard

Why:

The battery supplies power to the laptop when it is not connected to the charger.

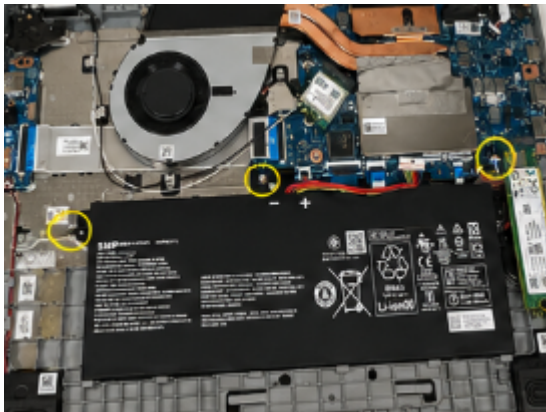


Step 5— Check all connections

Checking all internal connections before closing the laptop

Why:

Checking the connections before closing the laptop helps prevent problems during startup.



Step 6 — Replace the bottom cover

Replacing and securing the laptop's bottom cover

Why:

The bottom cover protects the internal components from dust, accidental contact and physical damage.



Step 7 — Power on and test the laptop

Powering on the laptop and checking its operation

Why:

Testing confirms that the components and connectors were reassembled correctly.



TASK-4

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Tool	Why it is important
1.Philips screwdriver	Used to remove and tighten most PC screws.
2.Small container for screws	Prevents screws from getting lost during disassembly.
3.Anti-static wrist strap	Helps reduce the risk of electrostatic discharge damaging components.
4.Clean workspace	Provides enough space to safely place components and screws.
5.Soft, clean surface	Protects components from scratches and accidental damage.
6.Flashlight	Helps see cables, screws and connectors inside the cabinet.